

Basit Ali - Principal Big Data Engineer

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SUMMARY

Principal Data Engineer with 9+ years of experience designing and leading large-scale data platforms across healthcare, telecom, government, retail, and IoT sectors. Skilled in building secure, scalable data architectures, real-time processing systems, and cloud-based solutions that improve performance and reduce costs. Recognized for leading teams, improving engineering standards, and ensuring data solutions deliver clear value across the organization.

CORE ARCHITECTURE COMPETENCIES

- Data Lake, Lakehouse, Data Mesh, Data Fabric & Medallion Architecture
- Cloud Data Architecture, Modeling, Governance & Metadata Management
- ETL/ELT Architecture, Automation & Real-Time Streaming
- Data Integration, Federation, Virtualization & Master Data Management
- AI/ML & Generative AI-Enabled Data Platform Architecture

TECHNICAL SKILLS

1. **Big Data & Analytics:** Cloudera Data Platform | Apache Spark | Kafka | Airflow | Databricks | Apache Hudi | Hadoop | HDFS | MapReduce | Neo4j | Hive | Power Apps
2. **Real-Time Streaming:** Apache Kafka | Kafka Streams | Spark Streaming | Apache Flink | Apache NiFi | AWS Kinesis Firehose | Event-Driven Architecture
3. **Cloud Platforms:** Azure (ADF, Synapse Analytics, Databricks, Functions, DevOps) | AWS (EMR, Glue, Lambda, Kinesis, Redshift, S3, Step Functions) | GCP | Microsoft Fabric
4. **Programming:** Python | Java | SQL | Node.js | Angular | Spark
5. **Databases:** HP Vertica | Teradata | Redshift | Snowflake | BigQuery | Azure Synapse | MongoDB | Elasticsearch | Redis | PostgreSQL
6. **DevOps & Tools:** GitHub | Azure DevOps | Jenkins | TFS | Trello | Jira | MS Office | MS Visio
7. **Artificial Intelligence:** Generative AI | RAG Applications | Agentic AI | Deep Learning | Machine Learning

WORK EXPERIENCE

Lean Business Services, Riyadh Saudi Arabia

February 2025 – Present

Senior Big Data Engineer

- Architected and implemented a large-scale Patient Profile Platform using Apache Spark, consolidating and centralizing healthcare data from hospitals and medical facilities across the Kingdom into a unified enterprise DataLake — establishing a single source of truth for millions of patient records at national scale.
- Designed and developed patient linkage profiling algorithms to match and merge patient identities across multiple disparate hospital systems, eliminating data duplication and enabling a complete 360-degree longitudinal patient view across facilities.
- Built and managed end-to-end data pipelines responsible for ingesting, transforming, and loading structured and unstructured clinical data from various source systems including lab results, radiology, pharmacy, and outpatient records into the centralized Data Lake.
- Developed advanced analytics and insight-generation layers on top of the centralized Data Lake, enabling stakeholders to derive actionable population health insights, clinical trends, and operational performance metrics from unified patient data.
- Engineered a metadata-driven ingestion framework using Streamlit and Python to automate data onboarding, enabling configuration of SQL, MySQL, Oracle, and NoSQL sources without manual coding — significantly accelerating data delivery and reducing onboarding effort across new hospital integrations.
- Architected real-time data processing pipelines using Spark Streaming, Kafka, and Apache NiFi to ingest high-volume healthcare data streams, achieving sub-5-second data availability for critical operational and clinical decision-making.
- Optimized long-running Oracle extraction jobs and batch processing workflows, reducing processing windows by 40% and improving reliability and predictability of daily data loads across all integrated facilities.

- Implemented strong data governance, quality control frameworks, and privacy safeguards across all pipelines, ensuring compliance with healthcare data protection regulations and national data security standards.
- Reduced infrastructure costs by ~30% by optimizing event-driven architectures, eliminating redundant pipelines, and decommissioning inefficient legacy workflows — improving overall system performance and scalability.

Ai-Elements, Riyadh Saudi Arabia

September 2023 – January 2025

Senior Big Data Engineer

- Designed and implemented multi-source data pipelines using Apache Spark, NiFi, and Airflow, ingesting structured and semi-structured data from 10+ enterprise systems into a centralized analytics platform, enabling near real-time operational insights for business stakeholders.
- Collaborated with data science and product teams to translate business requirements into Airflow-orchestrated data workflows, reducing manual processing effort by 25% and accelerating time-to-insight for key decision-makers.
- Led development and maintenance of end-to-end pipelines using Airflow, NiFi, and Spark, enabling real-time ETL from diverse systems to target platforms.
- Translated business needs into scalable pipeline solutions while coordinating with stakeholders to gather, refine, and document requirements.
- Managed and optimized on-premises Cloudera/Hadoop data workflows handling terabyte-scale batch processing, achieving 99.9% pipeline reliability and full compliance with organizational data security standards.
- Standardized engineering best practices including automated pipeline monitoring, code review processes, and data quality checks, reducing production incidents by 20% and improving system resilience across the on-premises data infrastructure.
- Applied Clean Code practices and refactored legacy codebases, reducing technical debt by 30% and improving maintainability.
- Built a Cloudera-based PoC for AI-driven insights using Spark and Hadoop to process large datasets, train ML models, and enhance performance.

Systems Limited, Islamabad Pakistan

July 2019 – September 2023

Senior Big Data Engineer

- Designed and delivered Microsoft Fabric Medallion Architecture and scalable data pipelines using Apache Spark, Hadoop, and Big Data ecosystems, improving ingestion, transformation, and analytics performance.
- Built distributed ETL frameworks with Spark and Python on AWS EMR, processing 10+ TB of data daily, improving data accuracy by 35%, and enabling actionable insights across telecom, retail, and government clients.
- Implemented strong data governance, privacy controls, and end-to-end documentation, ensuring compliance and operational consistency.
- Engineered AWS-based data platforms using EMR, Glue, Lambda, Kinesis, Step Functions, CloudFormation, RDS, S3, and Redshift, enabling both batch and real-time processing at scale.
- Developed real-time ingestion pipelines using Kinesis Firehose, incremental processing with Apache Hudi, and automated orchestration using AWS Lambda and Step Functions.
- Architected and delivered an end-to-end real-time streaming data processing platform using Apache Kafka, Spark Streaming, and Apache Flink, enabling sub-second latency event ingestion and processing for high-volume data streams across telecom and retail domains — supporting live analytics, fraud detection, and operational monitoring use cases.
- Built multi-cloud pipelines integrating Azure and AWS, including ADF, Azure Functions, Queues, Redis, and Databricks for near real-time ETL and cross-platform data movement.
- Applied Clean Code, SOLID principles, and TDD, refactoring legacy systems and reducing technical debt by 30% while improving code quality and maintainability.
- Implemented CI/CD automation using Jenkins, integrating build pipelines with AWS to orchestrate Spark jobs, data workflows, and infrastructure deployments.
- Designed and optimized SQL-based data validation, creating stored procedures and automating test routines to support QA teams during data verification.

Metis International Private Limited, Islamabad Pakistan

February 2018 – June 2019

Big Data Developer

- Developed graphical representations of data using Neo4j, enhancing data visualization capabilities.

- Designed and implemented a comprehensive web-based Big Data portal using Python and Django, facilitating efficient data management and analysis.

Big Data Developer (Dubai Future Accelerators - Onsite)

- Conducted social media analysis to gain insights into crime perspectives, hate speech, and sentiment analysis using Python.
- Utilized Python for Social Media Analysis on Drugs, identifying the location of users posting content related to the purchase or sale of drugs on social media platforms, supporting the Anti-Narcotics Department.

Front End Developer

- Employed Angular 2 and Angular 5 technologies to develop an interactive front-end application.
- Created real-time alert systems, live vehicle tracking features, geofencing functionality, and analytical dashboards for monitoring fuel consumption, temperature changes, speed, and driving behavior.

CERTIFICATIONS AND TRAININGS

- Microsoft Certified Azure Data Engineer Associate
- Cloudera Technical Professional (CTP) Accreditation
- Cloudera Technical Specialist (CTS) Accreditation
- Cloudera Technical Expert (CTE) Accreditation
- Future Leadership Program offered by Lahore University of Management Sciences (LUMS)

EDUCATION

International Islamic University, Islamabad Pakistan
Bachelor of Science in Software Engineering

February 2018

ACHIEVEMENTS

- Team of the Year (2021) and Star Performer (2020, 2021) – Systems Limited.
- Lead and mentor a high-performing Big Data engineering team.
- Completed the Future Leadership Program – LUMS (REDC).